

# Quantum error mitigations for quantum approximate optimization algorithms on IBM quantum processors

S. Liu, K. Yu

Submitted to the Quantum Communications and Quantum Imaging XXII Conference  
to be held at San Diego, United States  
August 18 - 23, 2024

Computational Science Initiative  
**Brookhaven National Laboratory**

**U.S. Department of Energy**  
USDOE Office of Science (SC), Advanced Scientific Computing Research (ASCR)

Notice: This manuscript has been authored by employees of Brookhaven Science Associates, LLC under Contract No. DE-SC0012704 with the U.S. Department of Energy. The publisher by accepting the manuscript for publication acknowledges that the United States Government retains a non-exclusive, paid-up, irrevocable, world-wide license to publish or reproduce the published form of this manuscript, or allow others to do so, for United States Government purposes.

## **DISCLAIMER**

This report was prepared as an account of work sponsored by an agency of the United States Government. Neither the United States Government nor any agency thereof, nor any of their employees, nor any of their contractors, subcontractors, or their employees, makes any warranty, express or implied, or assumes any legal liability or responsibility for the accuracy, completeness, or any third party's use or the results of such use of any information, apparatus, product, or process disclosed, or represents that its use would not infringe privately owned rights. Reference herein to any specific commercial product, process, or service by trade name, trademark, manufacturer, or otherwise, does not necessarily constitute or imply its endorsement, recommendation, or favoring by the United States Government or any agency thereof or its contractors or subcontractors. The views and opinions of authors expressed herein do not necessarily state or reflect those of the United States Government or any agency thereof.

# Quantum error mitigations for quantum approximate optimization algorithms on IBM quantum processors

Sixian Liu<sup>a</sup>, Hyunkyung Lim<sup>a</sup>, Jia Choi<sup>b</sup>, Paulo Castillo<sup>c</sup>, Gilchan Park<sup>d</sup>, and Kwangmin Yu<sup>d,\*</sup>

<sup>a</sup>Department of Applied Mathematics and Statistics, Stony Brook University, Stony Brook, New York, 11794, USA

<sup>b</sup>Jericho High School, Jericho, NY 11753, USA

<sup>c</sup>Electrical and Computer Engineering, Farmingdale State College, Farmingdale, New York 11735, USA

<sup>d</sup>Computational Science Initiative, Brookhaven National Laboratory, Upton, New York 11973, USA

## ABSTRACT

In recent years, there have been significant advancements in various aspects of quantum computing. However, despite this substantial progress, the availability of fault-tolerant quantum computers is still out of reach and may remain so for decades. Therefore, a key challenge is to leverage current NISQ devices to achieve a quantum advantage effectively. In this context, the Quantum Approximate Optimization Algorithm (QAOA) was proposed to potentially demonstrate computational advantages in combinatorial optimization problems using NISQ computers. Meanwhile, quantum error mitigation (QEM) techniques have been developed to address errors, with their effectiveness validated in practical problems involving more than 100 qubits. Therefore, in this paper, we optimize QAOA circuits and apply various error mitigation methods, such as dynamic decoupling and Pauli-twirling, to scale problem sizes on IBM quantum processors. Additionally, we discuss optimal implementation strategies for scalable QAOA. We test our implementations on Max-Cut problems and compare our results with previous works.

**Keywords:** Quantum Optimization, QAOA, Maxcut Problem, IBM Quantum

## 1. INTRODUCTION

Although quantum computers have achieved significant milestones in recent years, Noisy-Intermediate Scale Quantum (NISQ)<sup>1</sup> computers continue to grapple with noise and errors. These issues include measurement errors, multi-qubit gate errors, and, most critically, short decoherence times. The broad applications of quantum computers are currently restricted by inadequate error tolerance and decoherence. Therefore, modern quantum computers are not only far from achieving quantum supremacy in practical applications, but also face challenges in implementing conventional quantum algorithms, such as number factoring, unstructured searches, and solving linear systems of equations. Furthermore, fault-tolerant quantum computers are presently out of reach and may continue to be so for decades. Consequently, it is crucial to use contemporary quantum computers efficiently to address practical problems.

In this vein, various quantum error mitigation methods<sup>2</sup> are developed and applied<sup>3-5</sup> to Hamiltonian simulations, as well as utilizing pulse level optimization.<sup>6</sup> Additionally, quantum noise-resilient algorithms, such as Quantum Amplitude Estimation (QAE)<sup>7-14</sup> and Grover's search,<sup>15-18</sup> were developed to run the traditional quantum algorithms on noisy quantum computers. On the other hand, Variational Quantum Algorithms (VQAs), such as Quantum Approximate Optimization Algorithm (QAOA),<sup>19,20</sup> Variation Quantum Eigensolver (VQE),<sup>21</sup> and Quantum Machine Learning,<sup>22-26</sup> have drawn attention as a promising candidate to achieve quantum advantage on NISQ devices. These variational quantum algorithms work in conjunction with classical computers. Since the quantum circuit parameters are tuned by classical computation in VQAs, they are iterative. Since

---

\* : Send correspondence to Kwangmin Yu, E-mail : kyu@bnl.gov

these algorithms use classical computers in the process, they undermine the desired quantum speed-up, despite having better accuracy and gate fidelity due to the shorter circuit depth. As a result of VQAs having a short circuit depth, this helps to mitigate quantum noise.

QAOA is a widely studied VQA, which was introduced in 2014 by *Farhi et al.*,<sup>19</sup> to solve combinatorial optimization problems of finite dimension on gate-model quantum devices. QAOA reformulates the optimization problem into an approximation problem of finding the ground state of a Hamiltonian with a variational ansatz. The ansatzes are parameters of the quantum gates of the QAOA and they are used to determine the solution through the use of the classical computers. This process is repeated until the solution reaches a threshold criteria. The QAOA is considered a discretized time evolution of the quantum annealing process. When the annealing time goes to infinity, the adiabatic theorem<sup>27</sup> guarantees the quantum annealing process converges to the ground states if the adiabatic conditions are satisfied. However, when the time discretization number,  $p$ , is finite, the convergence of QAOA is not guaranteed.

In the practical implementation of QAOAs on quantum devices, we have a better accuracy as we increase  $p$ . However, as we increase  $p$ , the circuit depth increases linearly. Hence, it also increases the quantum noise and error. Therefore, a large  $p$  experiences a trade-off between numerical error and device error when it is implemented on noisy quantum devices. In our previous work,<sup>28</sup> we studied the effect of the different  $p$  values and discussed optimal implementation methods for scalable QAOA and the accuracy of the implementation on IBM quantum processors. In this paper, we apply two quantum error mitigation (QEM) methods, such as Dynamical Decoupling (DD) and Pauli Twirling (PT), to the MaxCut problem and discuss the usability of the QEM methods on the QAOA problem.

## 2. MODEL FORMULATION

QAOA solves problems modeled using the Ising model, or equivalently, in the quadratic quantum unconstrained binary optimization (QUBO) form. The Ising model is a mathematical model for ferromagnetism in statistical mechanics. The energy Hamiltonian is formulated as:

$$H = - \sum_{i=1}^N h_i \sigma_i^z - \sum_{i<j}^N J_{i,j} \sigma_i^z \sigma_j^z$$

where  $\sigma_i^z$  denotes the Pauli  $z$ -matrix of site  $i$  with eigenvalues  $q_i \in \{-1, +1\}$ . Finding the ground state of the Hamiltonian is reformulated as follows:

$$f(\vec{q}) = - \sum_{i=1}^N h_i q_i - \sum_{i<j}^N J_{i,j} q_i q_j$$

where  $\vec{q} = (q_1, \dots, q_N)^T$  and  $q_i \in \{-1, +1\}$ .

On the other hand, Quadratic unconstrained binary optimization (QUBO) is a combinatorial optimization problem often studied in computer science due to its ability to model many problem types. In this problem, a cost function  $f$  is defined on an  $n$ -dimensional binary vector space  $\mathbb{B}^n$  onto  $\mathbb{R}$ .

$$f(\vec{x}) = \vec{x}^T Q \vec{x}$$

where  $Q$  is an upper diagonal matrix,  $\vec{x} = (x_1, \dots, x_N)^T$ , and  $x_i \in \{0, 1\}$ . The problem is finding the  $x^*$  which minimizes the cost function  $f$ . Since we have  $x_i^2 = x_i$ , the cost function is reformulated as follows:

$$f(\vec{x}) = \sum_{i=1}^N Q_{i,i} x_i + \sum_{i<j}^N Q_{i,j} x_i x_j.$$

In  $Q$ , the diagonal terms  $Q_{i,i}$  and the off-diagonal terms  $Q_{i,j}$  represent the linear terms and the quadratic terms, respectively. The unknowns of the Ising model  $s$  and the unknowns of the QUBO model  $x$  have the linear relation

$$q_i = 2x_i - 1 \quad \text{or} \quad x_i = \frac{1}{2}(q_i + 1).$$

Therefore, the two forms are interchangeable with the linear relation. In this study, we opt Ising model form to describe our combinatorial problem.

In this paper, we use the MaxCut problem to benchmark the QAOA implementation on IBM quantum devices, since the MaxCut problem is used in many experiments to benchmark the performance of the QAOA and its implementation. The QUBO modeling for the MaxCut in this paper follows the modeling in Ref.<sup>29</sup> as follows:

$$f(\vec{x}) = \sum_{(i,j) \in E} (2x_i x_j - x_i - x_j).$$

The QUBO form is converted to the Ising model as follows:

$$f(\vec{q}) = \sum_{(i,j) \in E} \left( \frac{1}{2} q_i q_j - \frac{1}{2} \right).$$

Since our goal is finding the  $\vec{q}$  that minimizes the cost function  $f(\vec{q})$ , we can remove the constant term,  $\frac{1}{2}$ . Finally, we have the following cost function to minimize as follows:

$$f(\vec{q}) = \sum_{(i,j) \in E} \frac{1}{2} q_i q_j. \quad (1)$$

where  $q_i \in \{-1, +1\}$ . For example, we have the cost function  $f(\vec{q}) = \frac{1}{2}(q_0 q_1 + q_0 q_2 + q_1 q_2 + q_2 q_3 + q_2 q_4 + q_3 q_4 + q_4 q_5 + q_5 q_6)$  for the MaxCut problem of the graph in Fig. 1.

### 3. EXPERIMENTAL SETTING

The QAOA is implemented to solve the combinatorial optimization MaxCut problem with linear topology. The simulation is set to run on the chain graph with 7 nodes. We select 7 qubits with relatively small errors and map each node of the chain graph to a qubit on the real quantum device. To reduce the decoherence error, we reduce the circuit depth to 5 ( $p = 1$ ) by running gates in parallel if possible.

We apply two quantum error mitigation techniques on IBM quantum devices, Dynamical Decoupling and Pauli Twirling, to achieve better results. For comparison, the QAOA is implemented with three settings: QAOA without quantum error mitigation, QAOA with dynamical decoupling, QAOA with dynamical decoupling and Pauli twirling combined. All three settings are run with the same initial variational parameters  $[1.195, 0.875]$  with gradient-free classical optimizer COBYLA on the same IBM quantum device `ibm_torino`. Initial variational parameters are chosen to be the optimal parameters from `qasm_simulator`. We explore the QAOA at shallow depths, specifically at  $p = 1$ , with the number of shots equal to 1024,  $8 \times 1024$ , and  $80 \times 1024$  for the three different settings above, respectively. The quantum-classical hybrid algorithm works as follows. The quantum circuit takes initial variational parameters and generates the output. Then, the classical optimizer updates variational parameters and loops back to the quantum circuit, outputting a new prediction of the energy expectation value. The optimal solution is obtained when the Hamiltonian cost function approaches the global minimum energy state.

Results are shown in histograms plotting the quasi-probability of each bit-string. The bit-string with the highest quasi-probability is probably the optimal solution to the MaxCut problems. We also compare the success rate for three different settings. The success rate is computed as follows. In the case the ratio of counts/shots is greater than the threshold, if the Hamiltonian cost function of the key is less than or equal to the global minimum, then it is an up success; otherwise, up fail. In the case the ratio of counts/shots is less than or equal to the threshold, if the Hamiltonian cost function of the key is greater than the global minimum, then it is a down success; otherwise, down fail. Both up success and down success count towards the success rate. The threshold is predefined so that the probability of each outcome is prominently greater than the uniform probability  $1/128 \approx 0.008$  while it is reasonable to compare across three settings. In particular, we define the threshold to be  $5/2^n$ , where  $n$  is the number of qubits assigned to a given problem. On histograms, we cut off the success rate at 0.04.

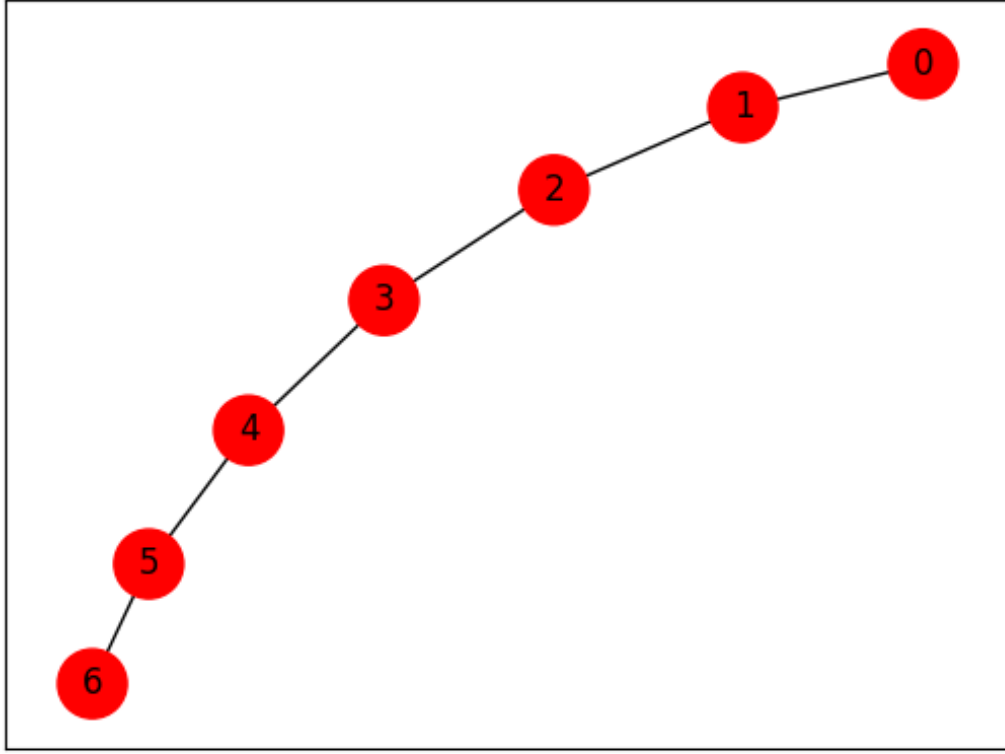


Figure 1: Chain Graph with 7 nodes.

### 3.1 Quantum Circuit Implementation

To implement Eq. (1) for the QAOA implementation, we start from the Ising coupling gate,  $R_{Z_i Z_j}(\theta)$  as follows:

$$R_{Z_i Z_j}(\theta) = \exp\left(-i\frac{\theta}{2}\sigma_i^z\sigma_j^z\right) = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 & 0 & 0 \\ 0 & e^{i\frac{\theta}{2}} & 0 & 0 \\ 0 & 0 & e^{i\frac{\theta}{2}} & 0 \\ 0 & 0 & 0 & e^{-i\frac{\theta}{2}} \end{pmatrix}$$

which is implemented as **RZZGate** in IBM Qiskit. The circuit diagram of the **RZZGate** is described as follows:

$$R_{ZZ}(\theta) = \begin{array}{c} \bullet \quad \bullet \\ \hline \oplus \quad \boxed{R_z(\theta)} \quad \oplus \end{array}$$

where we have  $R_z(\theta) = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix}$ .

### 3.2 Quantum Error Mitigations

The troublesome challenge of running quantum algorithms on contemporary quantum devices, including IBM Quantum processors, is the errors and noise on the quantum devices. To cope with the errors and noises, quantum error correction (QEC) was suggested.<sup>30,31</sup> However, QEC has a qubit overhead that is daunting to implement on a large problem on the contemporary quantum processors even though they are optimized.<sup>32,33</sup> On the other hand, quantum error mitigation (QEM) accepts the imperfection of contemporary quantum devices and adopts methods of mitigating or suppressing quantum errors and noises. QEM has a low or no qubit overhead. In recent years, various QEM methods have been developed, and their practicality has been proven in practical problems.<sup>4,5,34,35</sup> Hence, we apply two QEM methods, Pauli Twirling (PT) and Dynamical Decoupling (DD), to cope with the quantum device errors and noises in our experiments.

#### 3.2.1 Pauli Twirling

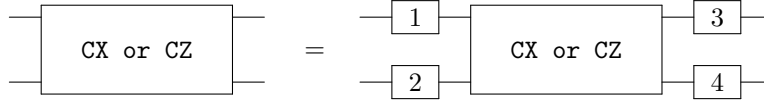


Figure 2: A circuit diagram for Pauli twirling for CX, CZ, or ECR gates. At positions 1, 2, 3, and 4, Pauli gates,  $\{I, \sigma^x, \sigma^y, \sigma^z\}$  are placed.

Pauli twirling is a method averaging out the off-diagonal coherent errors of the circuits in the Pauli basis,  $\{I, \sigma^x, \sigma^y, \sigma^z\}$ .<sup>36–38</sup> In Pauli twirling, a Clifford gate is surrounded by Pauli gates back and forth, which are mathematically identical to the Clifford gate. The efficiency is empirically proved in previous studies.<sup>4,34</sup> Figure 2 shows the Pauli twirling method we applied to this study. First, we searched all the combinations of Pauli gates that are mathematically identical up to the global phase with only the Clifford gate (CX or CZ gate depending on the target quantum device in our experiments). This is our Pauli twirling gates set. Since the Pauli gate set has four elements and we have four positions, the search domain is  $256 (= 4^4)$  cases. The trivial case is placing the identity gate in the position 1, 2, 3, and 4 in Fig. 2. One non-trivial case is putting  $\sigma^z, \sigma^x, \sigma^z$ , and  $\sigma^x$  at 1, 2, 3, and 4, respectively. We duplicated 10 copies of the base quantum circuit. After that, we randomly chose a Pauli twirling gate combination out of the prepared Pauli twirling gates set and applied them to the Clifford gate as described in Fig. 2.

#### 3.2.2 Dynamical Decoupling

Dynamical decoupling (DD) is a quantum error mitigation method that reduces errors caused by spectator qubits. DD is implemented by periodic sequences of instantaneous control pulses that average out the coupling with the system environment to approximately zero.<sup>39</sup> In particular, a set of single qubit operators are interleaved using basis transformation on idle qubits so that environmental contamination from other qubits is decoupled. Consequently, the coherence time of the circuit becomes longer. The efficiency of DD is empirically tested in various environments.<sup>4,34,35,40,41</sup> In this study, we added  $(t/4, \sigma^x, t/2, \sigma^x, t/4)$  sequence in every idling period through Qiskit `PassManager` where  $t$  is the idling time except for the two `XGate` pulse durations.

### 3.3 Circuit Implementation for Experiments and Post Processing

In this study, we used a 133-qubit IBM quantum processor, `ibm_torino`, which is an IBM Heron r1 processor. This processor has the basis gate set,  $\{CZ, I, RZ, SX, X\}$  where CZ is a controlled-Z gate,  $I \otimes |0\rangle\langle 0| + Z \otimes |0\rangle\langle 0|$  or  $Z \otimes |0\rangle\langle 0| + I \otimes |0\rangle\langle 0|$ . In the basis sets, CZ is the two-qubit gates and the other gates are one-qubit gates. Since we have  $Z = HXH$  gate identity, CZ can be converted from CX. In our circuit construction, we used CX gates regardless of the target devices. The Qiskit transpiler converts all other gates into the gates in the basis gate set depending on the target devices.

To construct the quantum circuit with the error mitigation methods described in Sec. 3.2, we first transpiled the quantum circuits of each iteration. The circuit implementation of the QAOA iteration is described in Sec. 3.1. During the transpiling process, the logical quantum gates are converted into sets in the basis gate set with circuit optimizations. We applied the highest optimization level during the transpiling.

After that, we applied the Pauli twirling by duplicating each circuit 10 times including the base circuit, and surrounded the Clifford gate with a randomly chosen Pauli twirling combination out of the prepared Pauli twirling gates set as described in Fig. 2. Finally, we applied the dynamical decoupling method to all those circuits. To execute each circuit, we used 8,192 shots (repeated circuit execution for the measurement sampling) in all cases. After gathering all the measurement results of the 10 circuit duplication for the Pauli twirling.

#### 4. RESULTS

Histograms in Fig. 3, Fig. 4, Fig. 5 show the quasi-probabilities of top 20 bit-strings from the 7-qubit quantum circuit with  $p = 1$  depth on ibm\_torino. Due to the different number of shots in the three settings, we calibrate to quasi-probability for comparison purposes. Fig. 3 shows the outcomes of the quantum circuits without quantum error mitigation methods, and the outcome with the highest probability is at the bit-string  $|0101010\rangle$  with the number of shots 1024. In this case, the cutoff of the success rate is at the 40-threshold, and the success rate is 0.9809523. Fig. 4 shows the outcome of the quantum circuit with dynamical decoupling, and the outcome with the highest probability is at the bit-string  $|1010101\rangle$  with number of shots to be  $8 * 1024$ . The cutoff of the success rate is 320-threshold, and the success rate is 0.9921875. Fig. 5 demonstrates the results from the quantum circuit with error mitigation methods of dynamical decoupling and Pauli twirling, and the outcome with the highest probability is at the bit-string  $|0101010\rangle$  with number of shots to be  $8 * 10 * 1024$ . The cutoff of the success rate is at the 3200 threshold, and the success rate is 1.0. We observed that the quantum circuit with dynamical decoupling and Pauli twirling has a higher success rate than the other two quantum circuits.

As in Table. 1, we also compare the quasi-probability differences between the optimal bit-string and non-optimal bit-string. In particular, we measure the difference between the second and the third bar in the histogram. Without quantum error mitigation methods, the difference is 0.046875. With dynamical decoupling, the difference is 0.009644. With dynamical decoupling and Pauli twirling, the difference is 0.022937. Moreover, we compare the sum of the first two bit-strings with the highest probability. With quantum error mitigation methods, the sum is 0.18359375; with dynamical decoupling, the sum is 0.11633301; with both dynamical decoupling and Pauli twirling, the sum is 0.12164306. As shown in the last column of Table. 1, the number of iterations of classical optimizer COBYLA are compared for three settings. The lower number of iterations indicates the hybrid algorithm converges to the optimal solution faster. The number of iterations is 33, 31, and 29, respectively.

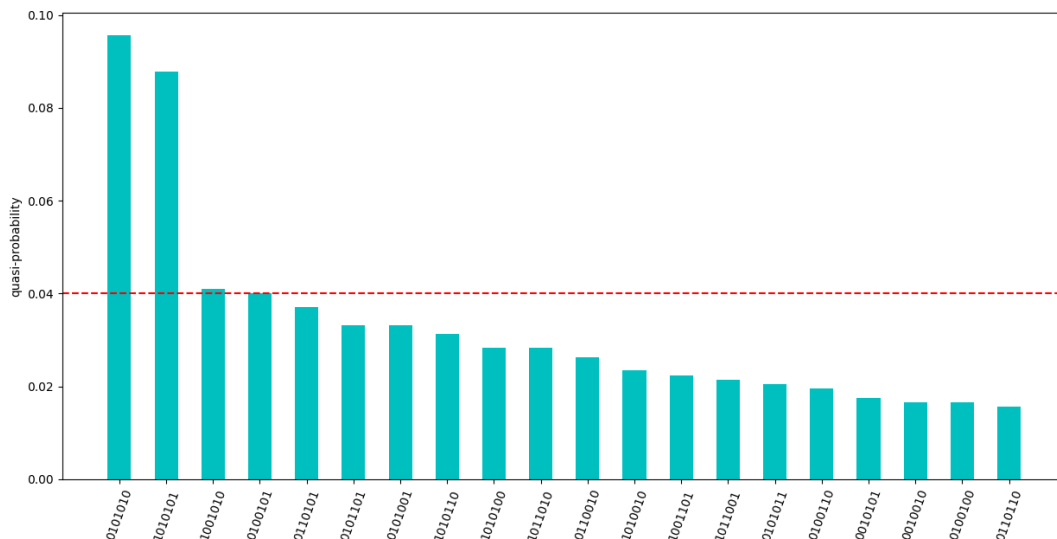


Figure 3: Results from ibm.torino without quantum error mitigation  $p = 1$ , number of shots = 1024. The success rate cutoff is at 0.04. The success rate is 0.9809523.



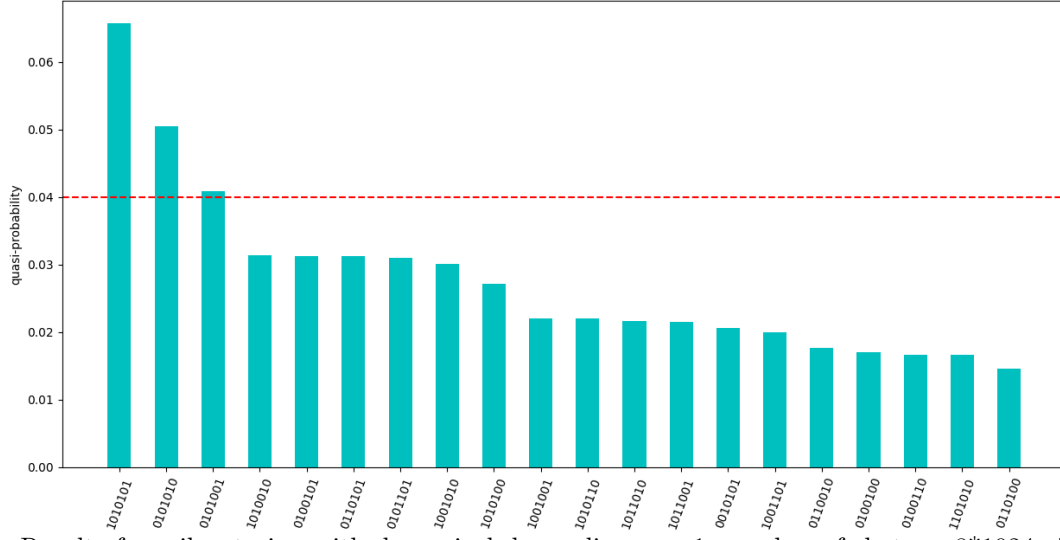


Figure 4: Results from `ibm_torino` with dynamical decoupling  $p = 1$ , number of shots =  $8 \cdot 1024$ . The success rate cutoff is at 0.04. The success rate is 0.9921875.

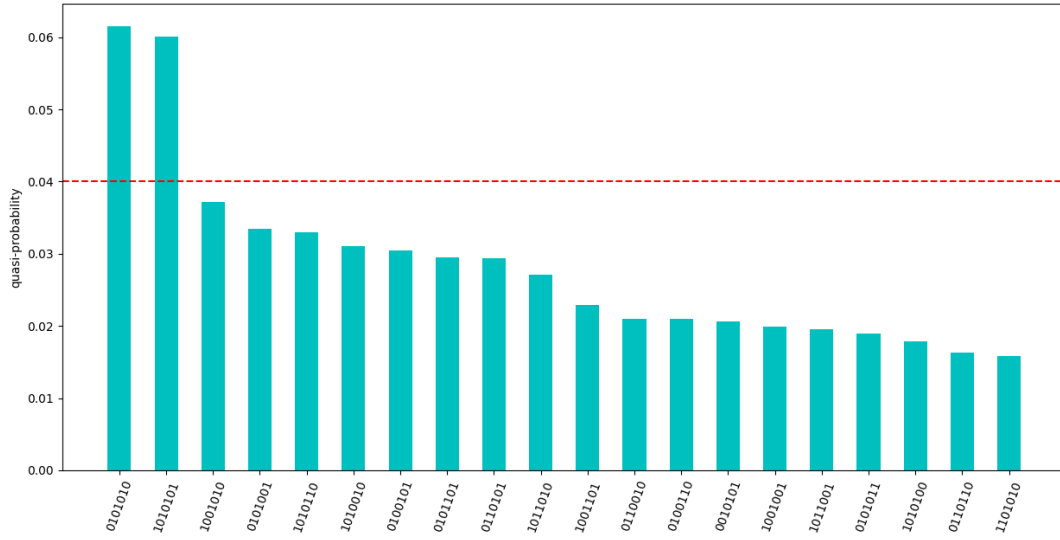


Figure 5: Results from `ibm_torino` with dynamical decoupling and Pauli twirling  $p = 1$ , number of shots =  $8 \cdot 10 \cdot 1024$ . The success rate cutoff is at 0.04. The success rate is 1.0.

| Method    | Success Rate | Difference | Sum        | Iterations |
|-----------|--------------|------------|------------|------------|
| No QEM    | 0.9809523    | 0.046875   | 0.18359375 | 33         |
| DD        | 0.9921875    | 0.009644   | 0.11633301 | 31         |
| DD and PT | 1.0          | 0.022937   | 0.12164306 | 29         |

Table 1: This table compares, among three settings (no quantum error mitigation, dynamical decoupling, dynamical decoupling, and Pauli twirling), success rate, the difference of quasi-probability between the second and third highest bit-strings, the sum of quasi-probability of first two highest bit-strings, and the number of iterations of classical optimizer COBYLA.

## 5. CONCLUSION

In this paper, we studied the quantum-classical hybrid QAOA on the combinatorial MaxCut problem for a 7-node chain graph. We implemented QAOA with three different settings: no quantum error mitigation, dynamical decoupling, and dynamical decoupling and Pauli twirling combined. We set the number of shots to be  $1024, 8 * 1024$  and  $8 * 10 * 1024$ , respectively, and chose the circuit depth  $p = 1$ .

Experiments show that QAOA with all three settings output bit-strings of optimal solutions. Regarding success rate, QAOA with quantum error mitigation methods outperform QAOA without quantum error mitigation. QAOA with dynamical decoupling and Pauli twirling has the highest success rate of 1.0. Moreover, QAOA with quantum error mitigation methods run with slightly fewer iterations of COBYLA. There is no supremacy of QAOA with quantum error mitigation on difference quasi-probability between optimal solutions and non-optimal solutions.

In our future work, we plan to enlarge the quantum circuit to 100 qubits and test QAOA with quantum error mitigation methods on the benchmark MaxCut problem. We will increase the circuit depth to  $p = 2, p = 3$  and perform on larger circuits for various types of graph structures. Implementations will be done via QISKIT, and experiments will be done on IBM quantum computers with more complex qubit connections.

## REFERENCES

- [1] Preskill, J., “Quantum Computing in the NISQ era and beyond,” *Quantum* **2**, 79 (Aug. 2018).
- [2] Cai, Z., Babbush, R., Benjamin, S. C., Endo, S., Huggins, W. J., Li, Y., McClean, J. R., and O’Brien, T. E., “Quantum error mitigation,” *Reviews of Modern Physics* **95**(4), 045005 (2023).
- [3] Kim, Y., Wood, C. J., Yoder, T. J., Merkel, S. T., Gambetta, J. M., Temme, K., and Kandala, A., “Scalable error mitigation for noisy quantum circuits produces competitive expectation values,” *Nature Physics* **19**, 752–759 (Feb. 2023).
- [4] Kim, Y., Eddins, A., Anand, S., Wei, K. X., van den Berg, E., Rosenblatt, S., Nayfeh, H., Wu, Y., Zaletel, M., Temme, K., and Kandala, A., “Evidence for the utility of quantum computing before fault tolerance,” *Nature* **618**, 500–505 (June 2023).
- [5] Chowdhury, T. A., Yu, K., Shamim, M. A., Kabir, M. L., and Sufian, R. S., “Enhancing quantum utility: Simulating large-scale quantum spin chains on superconducting quantum computers,” *Phys. Rev. Res.* **6**, 033107 (Jul 2024).
- [6] Zhang, K., Yu, K., Hao, K., and Korepin, V., “Optimal realization of yang–baxter gate on quantum computers,” *Advanced Quantum Technologies* **7**(4), 2300345 (2024).
- [7] Aaronson, S. and Rall, P., “Quantum approximate counting, simplified,” *Symposium on Simplicity in Algorithms*, 24–32 (Jan 2020).
- [8] Suzuki, Y., Uno, S., Raymond, R., Tanaka, T., Onodera, T., and Yamamoto, N., “Amplitude estimation without phase estimation,” *Quantum Information Processing* **19**(2), 75 (2020).
- [9] Yu, K., Lim, H., and Rao, P., “Practical numerical integration on NISQ devices,” in [*Quantum Information Science, Sensing, and Computation XII*], **11391**, 1 – 16, International Society for Optics and Photonics, SPIE (2020).
- [10] Yu, K., Lim, H., Rao, P., and Jin, D., “Comparison of amplitude estimation algorithms by implementation,” *arXiv preprint arXiv:2005.05300* (2020).
- [11] Rao, P., Yu, K., Lim, H., Jin, D., and Choi, D., “Quantum amplitude estimation algorithms on IBM quantum devices,” in [*Quantum Communications and Quantum Imaging XVIII*], **11507**, 49–60, SPIE (2020).
- [12] Rao, P., Yu, K., Lim, H., Jin, D., and Choi, D., “Amplitude estimation algorithms and implementations on noisy intermediate-scale quantum devices,” in [*Quantum 2.0*], QW6A–16, Optical Society of America (2020).
- [13] Grinko, D., Gacon, J., Zoufal, C., and Woerner, S., “Iterative quantum amplitude estimation,” *npj Quantum Information* **7**(1), 1–6 (2021).
- [14] Rao, P., Choi, S., and Yu, K., “Quantum multi-programming for maximum likelihood amplitude estimation,” in [*Quantum Computing, Communication, and Simulation IV*], **12911**, 126–134, SPIE (2024).
- [15] Zhang, K. and Korepin, V. E., “Depth optimization of quantum search algorithms beyond Grover’s algorithm,” *Physical Review A* **101**(3), 032346 (2020).
- [16] Zhang, K., Rao, P., Yu, K., Lim, H., and Korepin, V., “Implementation of efficient quantum search algorithms on NISQ computers,” *Quantum Information Processing* **20**(7), 1–27 (2021).
- [17] Park, G., Zhang, K., Yu, K., and Korepin, V., “Quantum multi-programming for grover’s search,” *Quantum Information Processing* **22**(1), 54 (2023).
- [18] Zhang, K., Yu, K., and Korepin, V., “Quantum search on noisy intermediate-scale quantum devices,” *Europhysics Letters* **140**, 18002 (sep 2022).
- [19] Farhi, E., Goldstone, J., and Gutmann, S., “A quantum approximate optimization algorithm,” *arXiv preprint arXiv:1411.4028* (2014).
- [20] Conley, R., Choi, D., Medwig, G., Mroczko, E., Wan, D., Castillo, P., and Yu, K., “Quantum optimization algorithm for solving elliptic boundary value problems on d-wave quantum annealing device,” in [*Quantum Computing, Communication, and Simulation III*], **12446**, 53–63, SPIE (2023).
- [21] Peruzzo, A., McClean, J., Shadbolt, P., Yung, M.-H., Zhou, X.-Q., Love, P. J., Aspuru-Guzik, A., and O’Brien, J. L., “A variational eigenvalue solver on a photonic quantum processor,” *Nature communications* **5**(1), 1–7 (2014).
- [22] Farhi, E. and Neven, H., “Classification with quantum neural networks on near term processors,” *arXiv preprint arXiv:1802.06002* (2018).

- [23] Altaisky, M., “Quantum neural network,” *arXiv preprint quant-ph/0107012* (2001).
- [24] Beer, K., Bondarenko, D., Farrelly, T., Osborne, T. J., Salzmann, R., Scheiermann, D., and Wolf, R., “Training deep quantum neural networks,” *Nature communications* **11**(1), 1–6 (2020).
- [25] Cong, I., Choi, S., and Lukin, M. D., “Quantum convolutional neural networks,” *Nature Physics* **15**(12), 1273–1278 (2019).
- [26] Baker, J. S., Park, G., Yu, K., Ghukasyan, A., Goktas, O., and Radha, S. K., “Parallel hybrid quantum-classical machine learning for kernelized time-series classification,” *Quantum Machine Intelligence* **6**(1), 18 (2024).
- [27] Born, M. and Fock, V., “Beweis des adiabatenatzes,” *Zeitschrift für Physik* **51**(3-4), 165–180 (1928).
- [28] Medwig, G., Choi, S., Castillo, P. C., and Yu, K., “Practical quantum approximate optimization algorithms on ibm quantum processors,” in [*Quantum Computing, Communication, and Simulation IV*], **12911**, 412–420, SPIE (2024).
- [29] Glover, F., Kochenberger, G., Hennig, R., and Du, Y., “Quantum bridge analytics i: a tutorial on formulating and using qubo models,” *Annals of Operations Research* **314**(1), 141–183 (2022).
- [30] Shor, P. W., “Scheme for reducing decoherence in quantum computer memory,” **52**, R2493–R2496 (Oct. 1995).
- [31] Calderbank, A. R. and Shor, P. W., “Good quantum error-correcting codes exist,” **54**, 1098–1105 (Aug. 1996).
- [32] Kivlichan, I. D., Gidney, C., Berry, D. W., Wiebe, N., McClean, J., Sun, W., Jiang, Z., Rubin, N., Fowler, A., Aspuru-Guzik, A., Neven, H., and Babbush, R., “Improved Fault-Tolerant Quantum Simulation of Condensed-Phase Correlated Electrons via Trotterization,” *Quantum* **4**, 296 (July 2020).
- [33] Lee, J., Berry, D. W., Gidney, C., Huggins, W. J., McClean, J. R., Wiebe, N., and Babbush, R., “Even More Efficient Quantum Computations of Chemistry Through Tensor Hypercontraction,” *PRX Quantum* **2**, 030305 (July 2021).
- [34] Kim, Y., Wood, C. J., Yoder, T. J., Merkel, S. T., Gambetta, J. M., Temme, K., and Kandala, A., “Scalable error mitigation for noisy quantum circuits produces competitive expectation values,” *Nature Phys.* **19**(5), 752–759 (2023).
- [35] Charles, C., Gustafson, E. J., Hardt, E., Herren, F., Hogan, N., Lamm, H., Starecheski, S., Van de Water, R. S., and Wagman, M. L., “Simulating  $\mathbb{Z}_2$  lattice gauge theory on a quantum computer.”
- [36] Bennett, C. H., Brassard, G., Popescu, S., Schumacher, B., Smolin, J. A., and Wootters, W. K., “Purification of Noisy Entanglement and Faithful Teleportation via Noisy Channels,” **76**, 722–725 (Jan. 1996).
- [37] Wallman, J. J. and Emerson, J., “Noise tailoring for scalable quantum computation via randomized compiling,” **94**, 052325 (Nov. 2016).
- [38] Cai, Z. and Benjamin, S. C., “Constructing smaller pauli twirling sets for arbitrary error channels,” *Scientific Reports* **9** (Aug. 2019).
- [39] Viola, L., Knill, E., and Lloyd, S., “Dynamical decoupling of open quantum systems,” *Phys. Rev. Lett.* **82**, 2417–2421 (1999).
- [40] Ezzell, N., Pokharel, B., Tewala, L., Quiroz, G., and Lidar, D. A., “Dynamical decoupling for superconducting qubits: A performance survey,” *Phys. Rev. Applied* **20**(6), 064027 (2023).
- [41] Niu, S. and Todri-Sanial, A., “Effects of Dynamical Decoupling and Pulse-Level Optimizations on IBM Quantum Computers,” *IEEE Trans. Quantum Eng.* **3**, 1–10 (2022).